

CAPSTONE PROJECT

Bike-Sharing-Demand-Prediction

PRESENTED BY

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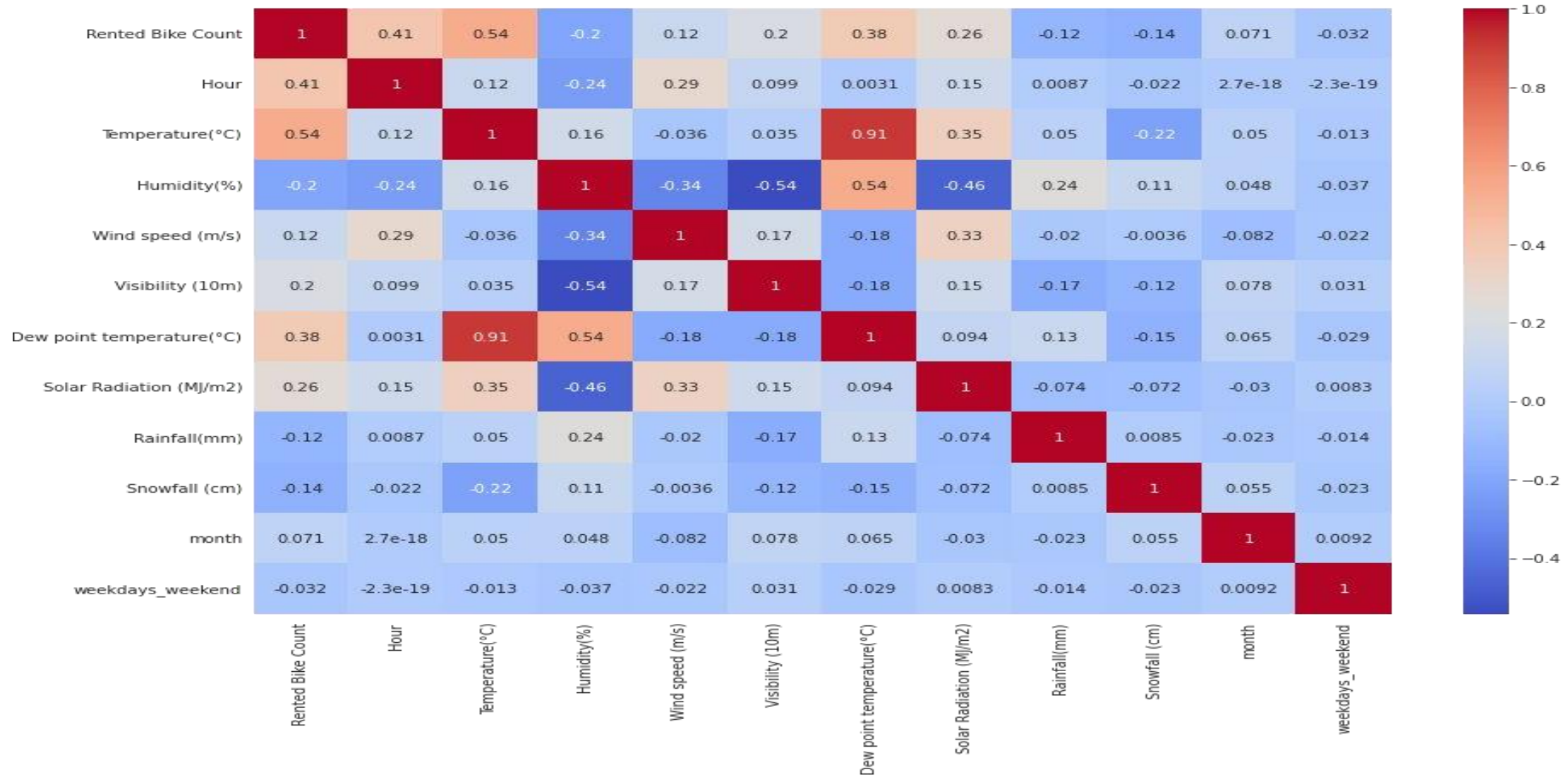
OUTLINE

- **Problem Statement** (Should not include solution)
- **Proposed System/Solution**
- **System Development Approach** (Technology Used)
- **Algorithm & Deployment**
- **Result (Output Image)**
- **Conclusion**
- **Future Scope**
- **References**

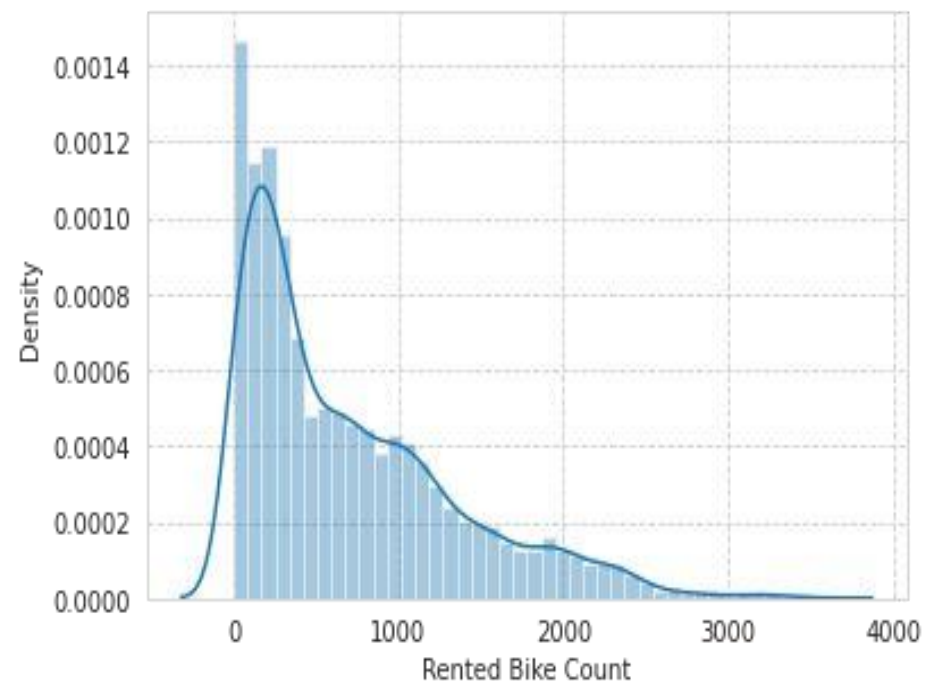
PROBLEM STATEMENT

Currently Rental bikes are introduced in many urban cities for the enhancement of mobility comfort. It is important to make the rental bike available and accessible to the public at the right time as it lessens the waiting time. Eventually, providing the city with a stable supply of rental bikes becomes a major concern. The crucial part is the prediction of bike count required at each hour for the stable supply of rental bikes.

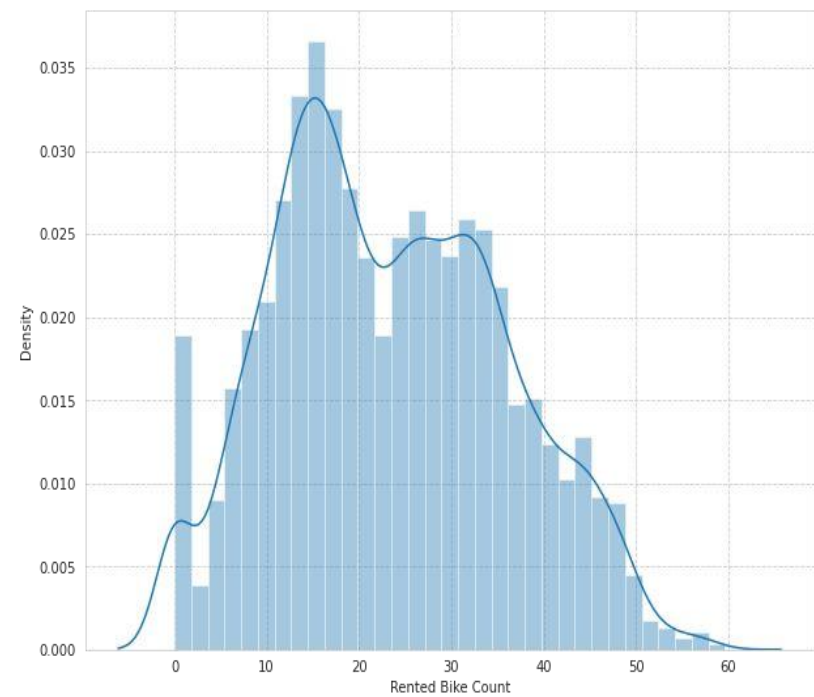
EDA - Feature Correlation



EDA (contd...)

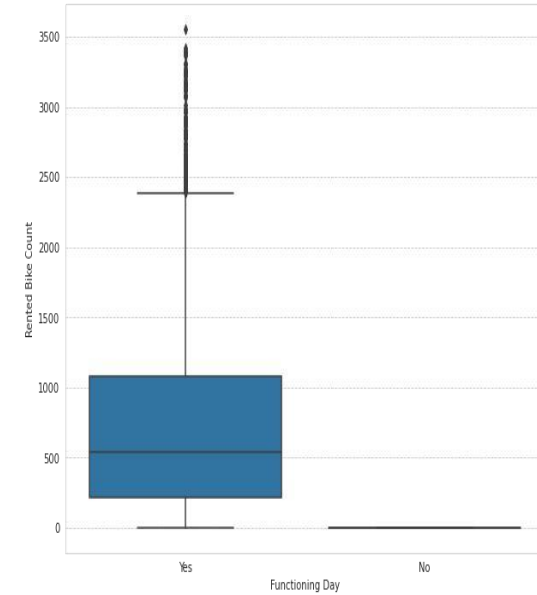
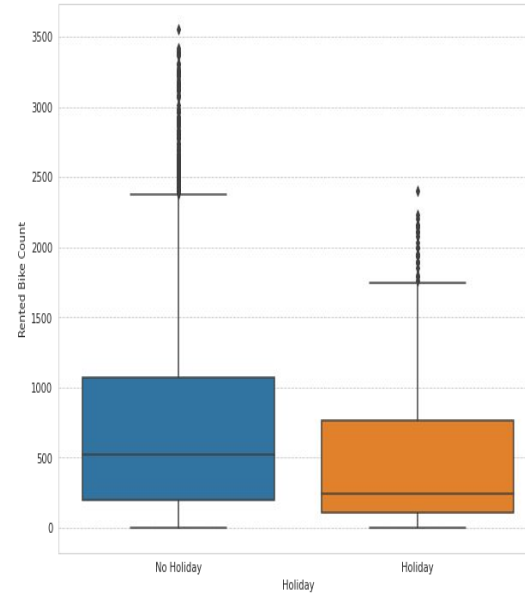
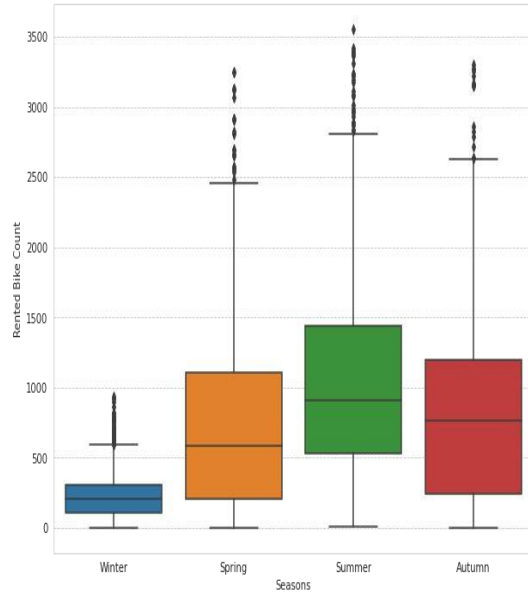


Distribution of rented bike count



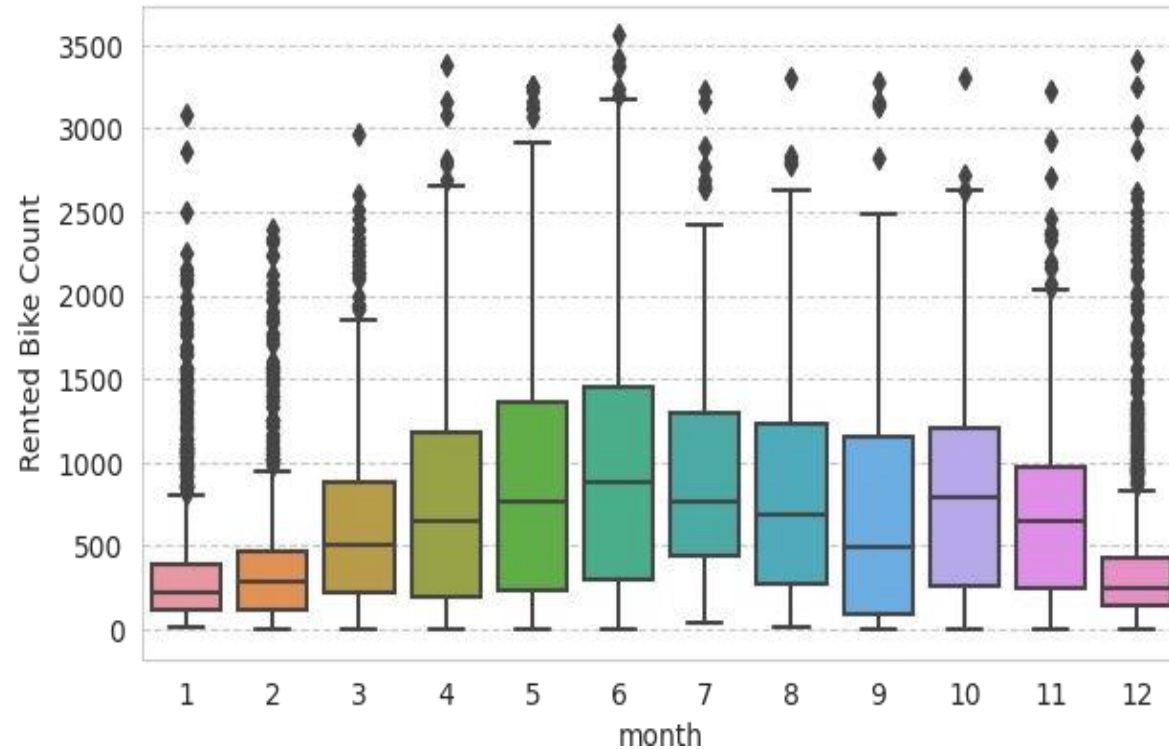
Square root transformation of rented bike count

EDA (contd...)



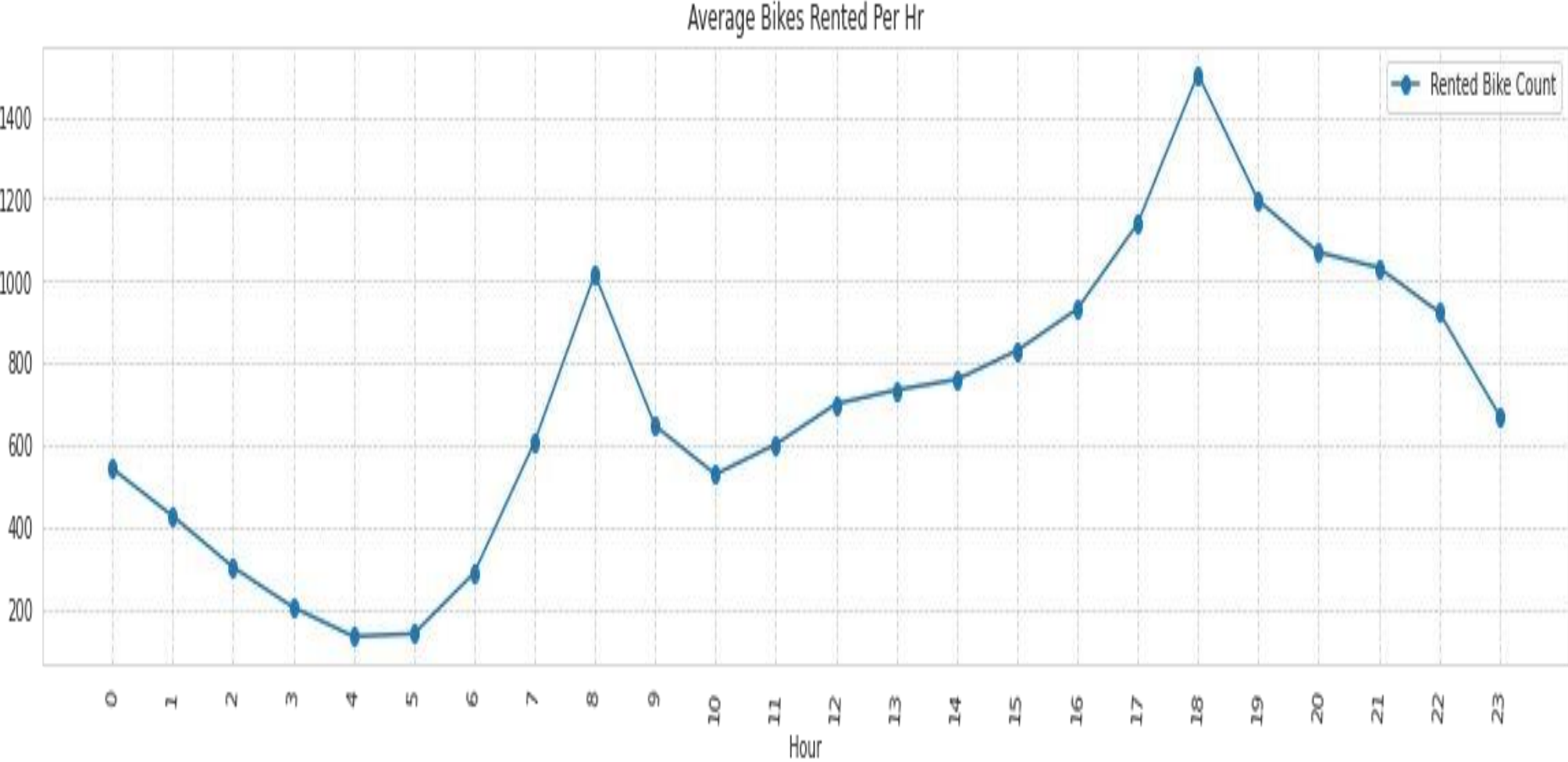
- Less demand on winter seasons
- Slightly Higher demand during Non holidays
- Almost no demand on non functioning day

EDA (contd...)



- We can see that there is less demand of Rented bike in the month of December, January, February i.e. during winter seasons
- Also demand of bike is maximum during May, June, July
- i.e. Summer seasons

EDA (contd...)



PROPOSED SOLUTION

- The proposed system aims to address the challenge of predicting the required bike count at each hour to ensure a stable supply of rental bikes. This involves leveraging data analytics and machine learning techniques to forecast demand patterns accurately. The solution will consist of the following components:
- **Data Collection:**
 - Gather historical data on bike rentals, including time, date, location, and other relevant factors.
 - Utilize real-time data sources, such as weather conditions, events, and holidays, to enhance prediction accuracy.
- **Data Preprocessing:**
 - Clean and preprocess the collected data to handle missing values, outliers, and inconsistencies.
 - Feature engineering to extract relevant features from the data that might impact bike demand.
- **Machine Learning Algorithm:**
 - Implement a machine learning algorithm, such as a time-series forecasting model (e.g., ARIMA, SARIMA, or LSTM), to predict bike counts based on historical patterns.
 - Consider incorporating other factors like weather conditions, day of the week, and special events to improve prediction accuracy.
- **Deployment:**
 - Develop a user-friendly interface or application that provides real-time predictions for bike counts at different hours.
 - Deploy the solution on a scalable and reliable platform, considering factors like server infrastructure, response time, and user accessibility.
- **Evaluation:**
 - Assess the model's performance using appropriate metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), or other relevant metrics.
 - Fine-tune the model based on feedback and continuous monitoring of prediction accuracy.
 - Result:

SYSTEM APPROACH

The "System Approach" section outlines the overall strategy and methodology for developing and implementing the rental bike prediction system. Here's a suggested structure for this section:

- System requirements
- Library required to build the model

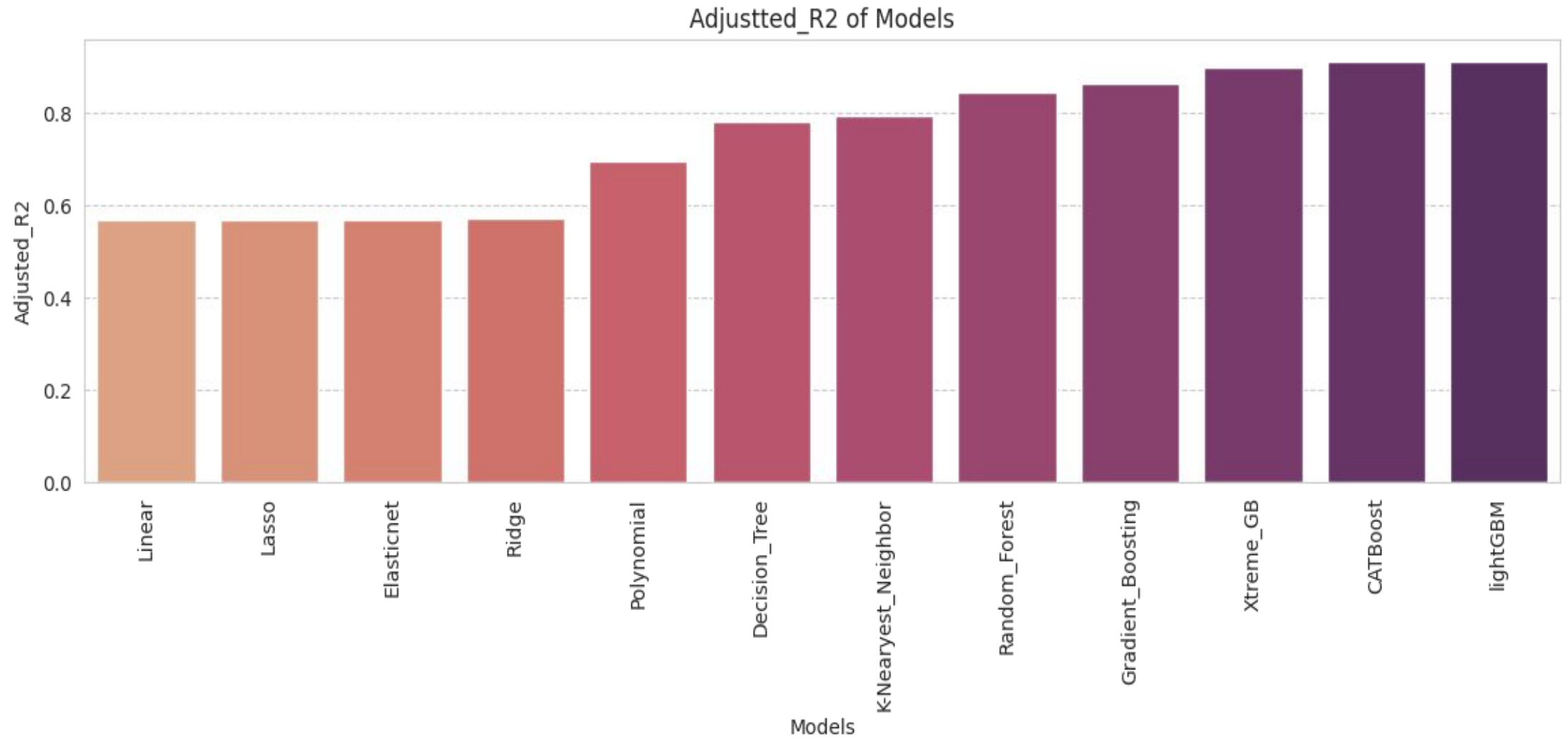
ALGORITHM & DEPLOYMENT

- In the Algorithm section, describe the machine learning algorithm chosen for predicting bike counts. Here's an example structure for this section:
- **Algorithm Selection:**
 - Provide a brief overview of the chosen algorithm (e.g., time-series forecasting model, like ARIMA or LSTM) and justify its selection based on the problem statement and data characteristics.
- **Data Input:**
 - Specify the input features used by the algorithm, such as historical bike rental data, weather conditions, day of the week, and any other relevant factors.
- **Training Process:**
 - Explain how the algorithm is trained using historical data. Highlight any specific considerations or techniques employed, such as cross-validation or hyperparameter tuning.
- **Prediction Process:**
 - Detail how the trained algorithm makes predictions for future bike counts. Discuss any real-time data inputs considered during the prediction phase.

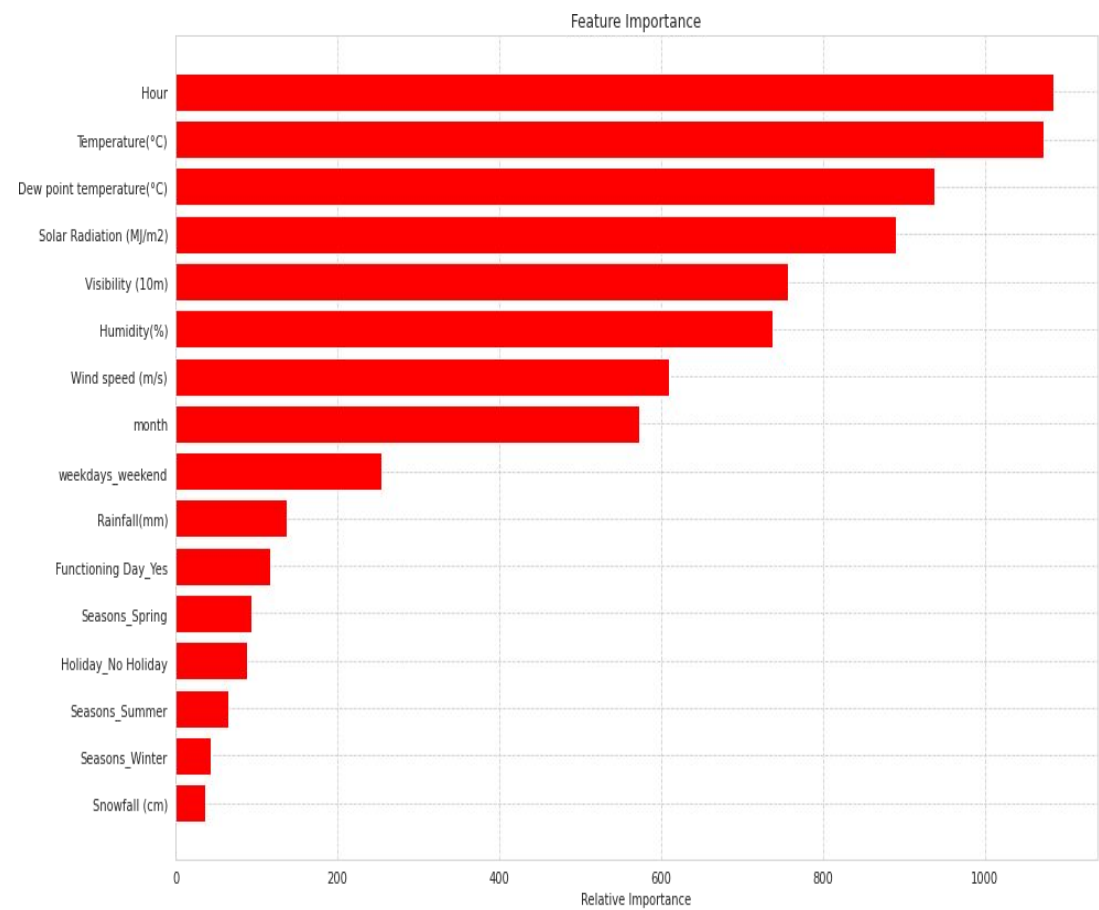
Model's Evaluation Matrices

	Models	Mean_square_error	Root_Mean_square_error	R2	Adjusted_R2
0	Linear	175590.552873	419.035264	0.572911	0.569766
1	Lasso	175560.907118	418.999889	0.572983	0.569839
2	Ridge	175248.935066	418.627442	0.573742	0.570603
3	Elasticnet	175346.867499	418.744394	0.573504	0.570363
4	Polynomial	123952.860328	352.069397	0.698509	0.696289
5	K-Nearyest_Neighbor	83411.759209	288.810940	0.796159	0.794659
6	Decision_Tree	88506.087215	297.499726	0.783710	0.782117
7	Random_Forest	62790.180423	250.579689	0.846554	0.845424
8	Gradient_Boosting	55090.172685	234.712958	0.865371	0.864380
9	Xtreme_GB	40812.801816	202.021785	0.900262	0.899528
10	CATBoost	36339.421527	190.629015	0.911194	0.910540
11	lightGBM	35410.753754	188.177453	0.913464	0.912826

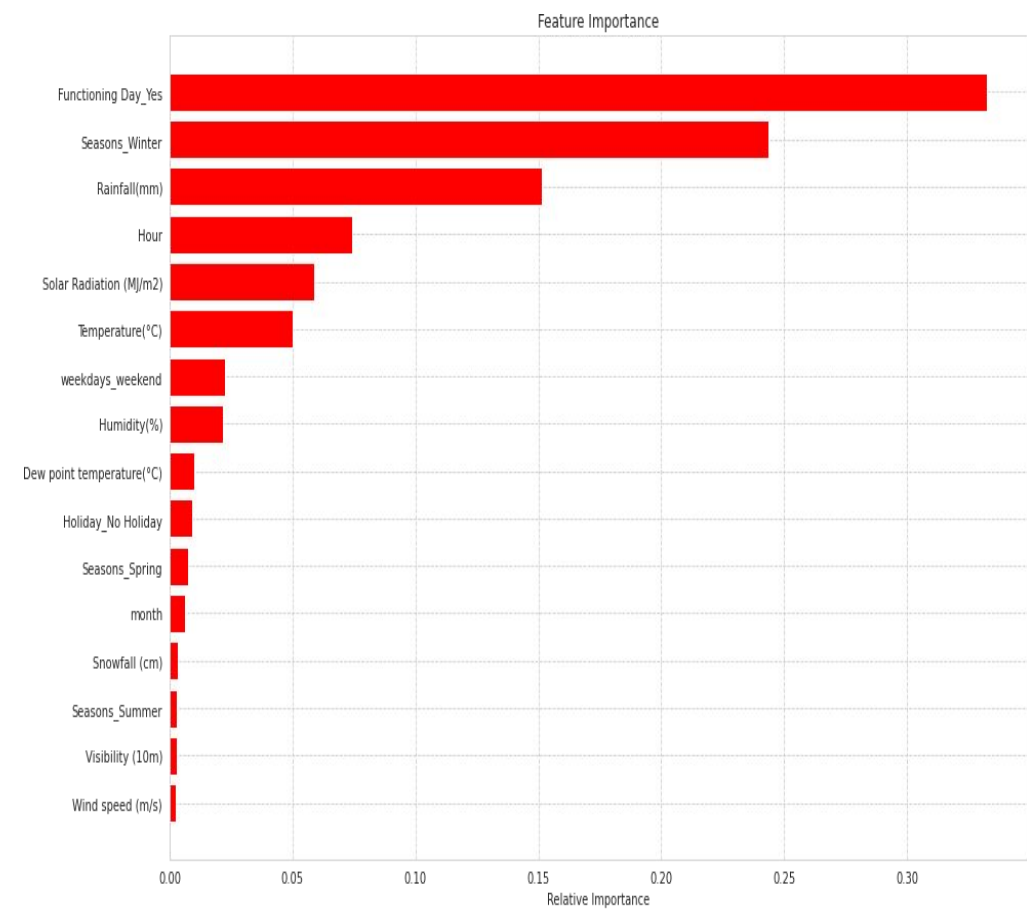
Adjusted R2 of Model's Performed



Feature Importance

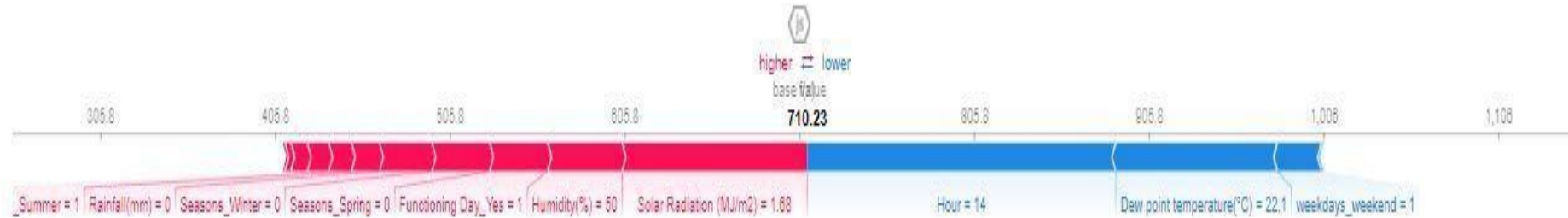


lightGBM



CatBoost

Model Explainability - SHAP



lightGBM



CatBoost

RESULT

Present the results of the machine learning model in terms of its accuracy and effectiveness in predicting bike counts. Include visualizations and comparisons between predicted and actual counts to highlight the model's performance.

CONCLUSION

Summarize the findings and discuss the effectiveness of the proposed solution. Highlight any challenges encountered during the implementation and potential improvements. Emphasize the importance of accurate bike count predictions for ensuring a stable supply of rental bikes in urban areas.

REFERENCES

List and cite relevant sources, research papers, and articles that were instrumental in developing the proposed solution. This could include academic papers on bike demand prediction, machine learning algorithms, and best practices in data preprocessing and model evaluation.

GitHub Link: https://github.com/Yuvaraj-3604/Bike_Sharing-Demand_Prediction.git

FUTURE SCOPE

Discuss potential enhancements and expansions for the system. This could include incorporating additional data sources, optimizing the algorithm for better performance, and expanding the system to cover multiple cities or regions. Consider the integration of emerging technologies such as edge computing or advanced machine learning techniques.

Thank you

